FREENESS CRITERION OVER A COMPLETE NOETHERIAN RING VIA MITTAG-LEFFLER AND UNIVERSAL INJECTIVITY

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Universal injectivity of the map into the completion, Mittag-Leffler property of the completion, and purity of the Mittag-Leffler property under a pure submodule. Same chapter flat.tex, labels lemma-universally-injective-to-completion-flat (proof line 2165, companion of S14-algebra-ui-to-completion) and lemma-lift-ML (proof line 2168); Algebra, label algebra-lemma-pure-submodule-ML (proof line 2171). Together they show that the module is Mittag-Leffler, since it is a pure submodule of its completion which is Mittag-Leffler.
- Projectivity of a countably generated flat Mittag-Leffler module. Algebra, label algebra-lemma-countgen-projective, invoked at proof line 2175. Converts the three properties obtained into projectivity over the base.

Source reference. Original lemma 059X: sources/stacks-chow/tags.txt, source lines 6830–6830. Navigation pointer for \ref{algebra-lemma-countgen-projective} used inside the counted span (line 2175); the argument that consumes it is retained in the span.

Source reference. Original lemma 059N: sources/stacks-chow/tags.txt, source lines 6821–6821. Navigation pointer for \ref{algebra-lemma-pure-submodule-ML} used inside the counted span (line 2171); the argument that consumes it is retained in the span.

Source reference. Original lemma 05DK: sources/stacks-flat/pinned-original-source.tex, source lines 2030–2030. Navigation pointer for \ref{lemma-lift-ML} used inside the counted span (line 2168); the argument that consumes it is retained in the span.

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Source reference. Original lemma 05DM: sources/stacks-flat/pinned-original-source.tex, source lines 2103–2103. Navigation pointer for \ref{lemma-universally-injective-to-completion-flat} used inside the counted span (line 2165); the argument that consumes it is retained in the span.

Lemma 0.1. *Let R be a ring. Let $I \subset R$ be an ideal. Let $R \to S$ be a ring map, and N an S-module. Assume*

(1) *R is Noetherian and I-adically complete,*
(2) *$R \to S$ is of finite type,*
(3) *N is a finite S-module,*
(4) *N is flat over R,*
(5) *N/IN is projective as a R/I-module, and*
(6) *for any prime $\mathfrak{q} \subset S$ which is an associated prime of $N \otimes_R \kappa(\mathfrak{p})$ where $\mathfrak{p} = R \cap \mathfrak{q}$ we have $IS + \mathfrak{q} \neq S$.*

Then N is projective as an R-module.

Proof. By Lemma 05DM the map $N \to N^\wedge$ is universally injective. By Lemma 05DK the module $N^\wedge$ is Mittag-Leffler. By Algebra, Lemma 059N we conclude that N is Mittag-Leffler. Hence N is countably generated, flat and Mittag-Leffler as an R-module, whence projective by Algebra, Lemma 059X. □